

VU Medical Zone

BIO 401 Final Term Spring 2021

Q1. How we draw slope of regression line? 2 Marks

We create a regression line by plotting two estimated values for y against their X component, then extending the line right and left.

Q2. What is estimate and estimator? 2 Marks

A statistic is also referred to as a sample statistic or, when estimating a parameter, a point estimator of a parameter. A specific value of a point estimator is referred to as a point estimate of a parameter.

Q3. Define SPSS. 2 Marks

Originally it is an acronym of Statistical Package for the Social Science but now it stands for Statistical Product and Service Solutions. One of the most popular statistical packages which can perform highly complex data manipulation and analysis with simple instructions.

Q4. Define Probability. 3 Marks

- It Link between sample and Population.
- Generalize results from sample to the population.
- The population is a theoretical and usually undefined quantity.
- Needed for statistical tests and confidence intervals.

Q5. Explain Regression Equation? 3 Marks

By using the least squares method (a procedure that minimizes the vertical deviations of plotted points surrounding a straight line) we are able to construct a best fitting straight line to the scatter diagram points and then formulate a regression equation in the form of:

$$\hat{y} = a + bX$$

$$\hat{y} = y + b(x - \bar{x})$$

$$b = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

Q6. Properties and uses of the Inter-Quartile Range?

5 Marks

Properties and uses of the Inter-Quartile Range

- o The IQR is generally used in conjunction with the median. Together, they are useful for characterizing the central location and spread of any frequency distribution, but particularly those that are skewed.
- o For a more complete characterization of a frequency distribution, the 1st and 3rd quartiles are sometimes used with the minimum value, the median, and the maximum value to produce a five number summary of the distribution.
- o Together these five numbers provide a good description of the centre, spread and shape of the distribution.

Q7. Specific mortality rate?

3 Marks

The cause-specific mortality rate is the mortality rate from a specified cause for a population. The numerator is the number of deaths attributed to a specific cause. The denominator remains the size of the population at the midpoint of the time period.

Q8. Additive rule dominant phenotype?

5 Marks

In this cross there are three events that can lead to a dominant phenotype.

- Two A gametes meet (giving AA genotype) OR
- A gamete from Mom meets with a gamete from Dad (giving Aa genotype) OR
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- a gamete from Mom meets with A gamete from Dad (giving Aa genotype).

Since each individual event has probability of occurrence = $\frac{1}{4}$

Q9. Four Windows of SPSS?

5 Marks

EFFORT BY AMAAN KHAN

The Four Windows: Data Editor

- ☐ Data Editor: Spreadsheet-like system for defining, entering, editing, and displaying data. Extension of the saved file will be “say.”
- ☐ Output Viewer: Displays output and errors. Extension of the saved file will be “spv.”
- ☐ Syntax Editor: Text editor for syntax composition. Extension of the saved file will be “sps.”
- ☐ Script Window: Provides the opportunity to write full-blown programs, in a BASIC-like language. Text editor for syntax composition. Extension of the saved file will be “sbs.”

Q10. Range Rule of Thumb? 2 Marks

The Range Rule of Thumb suggests that values are unusual if they are outside of these limits.

Maximum Usual Value = $\mu + 2\sigma$

Minimum Usual Value = $\mu - 2\sigma$

Q11. Three methods of Sampling? 5 Marks

Convenience sampling

- ☐ Convenience sampling is commonly known as unsystematic, accidental or opportunistic sampling. According to this procedure a sample is selected according to the convenience of the investigator.
- ☐ In this method of sampling the choice of sample items depends primarily on the judgment of the researcher. In other words, the researcher determines and includes those items in the sample which he thinks are most typical of the universe with regard to the characteristics of research project.
- ☐ A type of non probability sampling which involves the sample being drawn from that part of the population which is close to hand. That is, readily available and convenient.

QUOTA SAMPLING

- ☐ In this method, the sample size is determined first and then quota is fixed for various categories of population, which is followed while selecting the sample.
- ☐ In this method the quota has to be determined in advance and intimated to the investigator. The quota for each segment of the population may be fixed at random or with a specific basis. Normally such a sampling method does not ensure representativeness of the population.

SNOWBALL SAMPLING

- ☐ It refers to Identifying someone who meets the criteria for inclusion in the study.

- ☐ Selection of additional respondents is based on referrals from the initial respondents.

Q12. Negative value of correlation? 2 Marks

While if the sign is -ve this means an inverse or indirect relationship (which means an increase in one variable is associated with a decrease in the other).

Q13. Classical probability? 2 Marks

If an event can occur in N mutually exclusive and equally likely ways, and if m of these possess a trait E , the probability of the occurrence of E is equal to m/N .

$$P(E) = \frac{m}{N}$$

For an experiment consisting of n outcomes, The Classical definition of the probability assigns probability $1/n$ to each outcome or simple event.

Q14. Cluster and Quota sampling? 3 Marks

Cluster sampling

Divide population into clusters.

Randomly sample clusters.

Measure all units within sampled clusters.

QUOTA SAMPLING

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